

Shuwen Sun

Khoury College of Computer Sciences

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Research Interests

My research focuses on understanding distributed system behaviors and building fast, efficient, and reliable systems. In my PhD work I adapted techniques from newer storage device capabilities to build new distributed storage systems. My work also touches semantics and consistency guarantees of distributed systems.

Prior to my current work, I have worked on research projects in datacenter networking and performance diagnosis for distributed systems.

Professional Experience

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|------------------|---|---|
| 9/2025 – Present | Research Scientist | <i>Meta, New York City, NY</i> |
| | <ul style="list-style-type: none">• Performing research on building infrastructure for AI. | |
| 9/2018 – 9/2025 | Graduate Research Assistant | <i>Northeastern University, Boston, MA</i> |
| | <ul style="list-style-type: none">• Designed and implemented a highly-efficient and (single-key) linearizable object store, ZStore, that leverages ZNS SSDs to achieve high performance.• Designed and implemented application network functions (TLS cert validator, transcoder, remote dependency resolution proxy) which offload user/application level functions to the network. | |
| 5/2023 – 8/2023 | PhD Research & SWE Intern | <i>Google Global Networking, New York City, NY</i> |
| | <ul style="list-style-type: none">• Research prototypes for the Google Global Networking team. | |
| 6/2022 – 8/2022 | Research Intern | <i>ThousandEyes (Part of Cisco) Research, San Francisco, CA</i> |
| | <ul style="list-style-type: none">• Performed on Internet measurement related to anomaly detection in HTTP timing. Part of the Internet Research team. | |
| 1/2018 – 7/2018 | Staff Engineer | <i>Hariri Institute for Computing, Boston, MA</i> |
| | <ul style="list-style-type: none">• Staff engineer within MOC working on tracing tools for Openstack and a resource management framework in production. | |

Education

- 9/2018 – 10/2025 **Northeastern University, Boston, Massachusetts**
Ph.D. in Computer Science
Thesis: *Towards an Efficient and Strongly-consistent Distributed Object Store*
Advisor: Prof. [Peter Desnoyers](#)
- 9/2015 – 5/2017 **Boston University, Boston, Massachusetts**
M.S. in Computer Science
- 9/2011 – 5/2015 **Sun Yat-sen University, Guangzhou, China**
B.Eng. in Software Engineering

Publication

- C5. **A Fast, Efficient, and Strongly-Consistent Object Store**
Sun, Shuwen, Khor, Isaac, Shin, Ji-Yong, and Desnoyers, Peter.
16th ACM Symposium on Cloud Computing (SoCC'25). SoCC'25
- C4. **A case for IO efficiency as a research metric for storage systems**
Sun, Shuwen, Khor, Isaac, Ahmadi, Sina, Ren, Xiang (Jenny), Shin, Ji-Yong, and Desnoyers, Peter.
Under submission
- C3. **Endpoint-defined In-Network Functions**
Sun, Shuwen and Choffnes, David.
Under submission
- C2. **Toward Flexible Auditing for In-Network Functionality**
Sun, Shuwen and Choffnes, David.
CoNEXT-SW '22
- C1. **FlexNet: Enabling Flexibility in Cloud Networks**
Yu, Da, Mai, Luo, Sun, Shuwen, Krieger, Orran, and Fonseca, Rodrigo.
Under submission

Professional Service

To the Discipline:

- 2026 • **Program Committee** — IEEE MSST
- 2026 • **Reviewer** — Computer Networks
- 2025 • **Program Committee** — ACM SoCC

- 2025 • **Reviewer** — Computer Networks
- 2025 • **Reviewer** — IEEE Internet of Things Journal 2025
- 2024 • **Reviewer** — Peer-to-Peer Networking and Applications 2024
- 2024 • **Reviewer** — IEEE TIFS 2024
- 2021 • **External Reviewer** — CCS 2021
- 2020 • **Organizing Volunteer** — SIGCOMM 2020 Hallway Sessions
- 2020 • **External Reviewer** — IMC 2020
- 2019 • **External Reviewer** — NSDI 2020
- Summer 2017 • **Layer 9 Scriber** — SIGCOMM 2017

To Northeastern University & Khoury College:

- 3/2025 • **Panelist** — 2025 Ph.D. Open House Graduate Student Panel
- 9/2023-1/2024 • **Organizing committee & Program committee & Session chair** — 2nd Khoury Systems Day
- 1/2023 • **General Chair & Session Chair** — 1st Khoury Systems Day
- 7/2021 • **Co-organizer & panelist** — Workshop on Completing PhD Course Requirements at Khoury
- 3/2021 • **Moderator** — 2021 Ph.D. Open House Graduate Student Panel
- 1/2021 • **Ph.D. Open House Co-organizer** — 2021 Ph.D. Open House
- 1/2021 – 05/2022 • **Organizing member** — Khoury Graduate Students Association
- 12/2020 • **Ph.D. Admission Volunteer** — 2021 Ph.D. Admission
- 3/2020 • **Panelist** — 2020 Ph.D. Open House Graduate Student Panel
- 9/2019 • **Panelist** — 2019 Ph.D. Orientation
- 3/2019 • **Letter writer** — Khoury College Naming Ceremony Thank you Gift
- 3/2019 • **Panelist** — 2019 Ph.D. Open House Graduate Student Panel
- Spring 2019 • **Co-organizer** — Systems and Networking Reading Group

Skill Sets

- **Programming:** C/C++, Rust, Python, Go, Bash
- **Async Programming:** C++ (Boost.Asio, Boost.Beast), Rust (async/await, Tokio, futures)
- **Tools:** SPDK, DPDK, SGX, eBPF, Docker, OpenTracing/OpenTelemetry
- **Networking:** Kernel bypassing, NFV, RDMA, RoCE, Datacenter networking, SDN
- **Storage:** NVMe-over-fabric, SPDK, SSD, Zoned NameSpace SSD

- **Systems:** Distributed systems, End-to-end request tracing, Cloud computing

Talks & Posters

Talks:

- 2/2025 “ZStore: A Fast, Efficient, and Strongly-Consistent Object Storage System with ZNS SSDs”
Shuwen Sun. Talk at *Khoury Software Day 2025*, Boston, MA
- 2/2025 “ZStore: A Fast, Efficient, and Strongly-Consistent Object Storage System with ZNS SSDs”
Shuwen Sun. Talk at *2025 New England Systems Day*, Boston, MA
- 1/2024 “A case for IO efficiency as a research metric for storage systems”
Shuwen Sun. Talk at *2nd Northeastern System Day*, Boston, MA
- 1/2023 “Toward Flexible Auditing for In-Network Functionality”
Shuwen Sun. Talk at *Student Workshop co-located with ACM CoNEXT 2022*, Rome, Italy
- 12/2022 “Toward Flexible Auditing for In-Network Functionality”
Shuwen Sun. Talk at *1st Northeastern System Day*, Boston, MA
- 10/2017 “PYTHIA: A Just-in-Time Instrumentation Framework for Debugging Distributed Systems.” Lily Sturmann, **Shuwen Jethro Sun**. Talk at *2017 MOC Annual Workshop*. Boston, MA

Posters:

- 3/2020 “How well does your network (function) function? Understanding Network Functions Under User-level Use Cases” **Shuwen Jethro Sun**, David Choffnes. Poster at *Khoury Ph.D. Open House*.
- 12/2017 “PYTHIA: A Cross-layer Just-in-Time Instrumentation Framework for Debugging Distributed Applications.” Lily Sturmann, **Shuwen Jethro Sun**, Raja Sambasivan, Orran Krieger, Peter Portante. Poster at *IV New England Networking and Systems Day (NENS’17)*. Boston, MA
- 10/2017 “PYTHIA: A Just-in-Time Instrumentation Framework for Debugging Distributed Systems.” Lily Sturmann, **Shuwen Jethro Sun**, Rajul Kumar, Vladimir Pchelin, Orran Krieger, Peter Portante, Raja Sambasivan. Poster at *2017 MOC Annual Workshop*. Boston, MA

Honors & Awards

Academic Honors:

- 1/2018 Northeastern University Graduate School Ph.D. Fellowship.
(Admitted to Ph.D. program in Computer Science at Northeastern University)

Travel Grant Awards:

- 3/2021 NSDI ’21 Conference Student Grant, USENIX
- 2/2020 NSDI ’20 Conference Student Grant, USENIX
- 8/2019 SIGCOMM ’19 Conference Student Grant, NSF

8/2017	SIGCOMM '17 Conference Student Grant, NSF
7/2016	ATC '16 and HotCloud '16 Conference Student Grant, USENIX
 Miscellaneous:	
2/2022	Invited participants of Google Network Research Summit
9/2014	Second-class Scholarship for Outstanding Students at Sun Yat-sen University (Top 10%).
4/2011	Recipient of independent recruitment for Sun Yat-sen University in 2011 (Top 6%, roughly 660 of 11,000 in China).

Teaching Experience

Spring 2025	Teaching Assistant, CS 3650 Computer Systems, Northeastern Course instructors: Peter Desnoyers.
Spring 2024	Head Teaching Assistant, CS 3650 Computer Systems, Northeastern Course instructors: Peter Desnoyers and Cheng Tan.
Fall 2023	Teaching Assistant, CS 5600 Computer Systems, Northeastern Course instructors: Peter Desnoyers.
Fall 2021	Teaching Assistant, CS 3700 Networks and Distributed Systems, Northeastern Course instructors: David Choffnes, Sakib Miazi, Christo Wilson.

Personal Trivia

Languages: English (professional proficiency), Mandarin (native)

Affiliations

- Khoury College of Computer Sciences, Northeastern University
- [MOC Alliance](#)

References

Peter Desnoyers, Associate Professor
Northeastern University
✉ pjd@ccs.neu.edu

Ji-Yong Shin, Assistant Professor
Northeastern University
✉ j.shin@northeastern.edu

Orran Krieger, Research Professor
Boston University
✉ okrieg@bu.edu

Vasilis Pappas, Tech Lead
Google Global Networking
✉ vasilis@google.com